Monitoring CO_2 sequestration with electromagnetics in the presence of steel-cased wells

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Introduction

Geologic storage of CO_2 involves injection into saline aquifers or depleated oil and gas reservoirs

→ changing pore fluids alters electrical conductivity

→ electomagnetics (EM) may be useful for monitoring

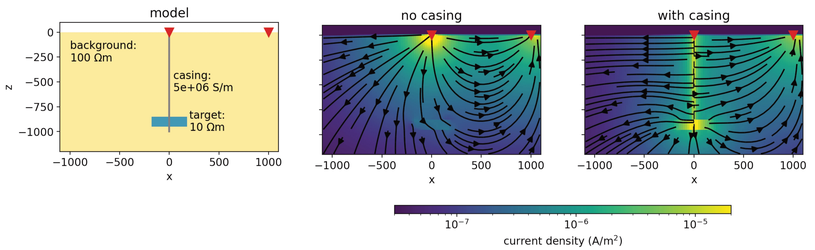
Steel cased wells are typically present

→ steel is highly conductive (5x10^6 S/m) and magnetic (50-150μ_0)

→ large impact on EM response: can help with detecting targets

→ EM can also be used to assess casing integrity

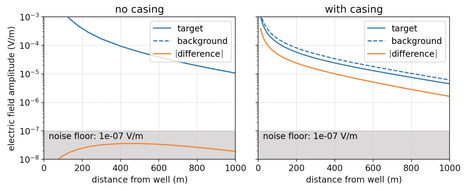
Steel casing can help excite deep targets



DC resistivity experiment

Currents channeled into conductive casing

Improves ability to detect deep targets



Numerical modelling

simpeg

Finite volume simulations

→ in time or frequency domain

Cylindrical meshes

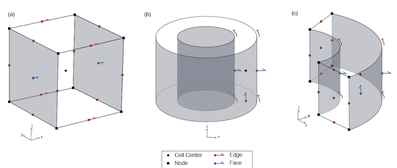
→ finely discretize casing

Open source software

→ enables re-use, adaptation

$$\nabla \times \vec{h} = \vec{j} \qquad \vec{j} = \sigma \vec{e}$$

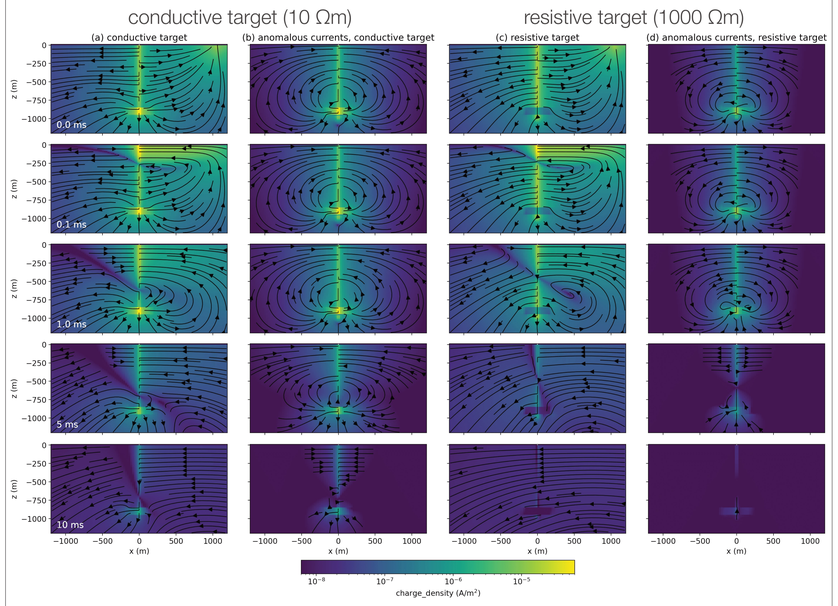
$$\nabla \times \vec{e} = -\frac{\partial \vec{b}}{\partial t} \qquad \vec{b} = \mu \vec{h}$$



Electromagnetics

Time-varying current. Step-off waveform.

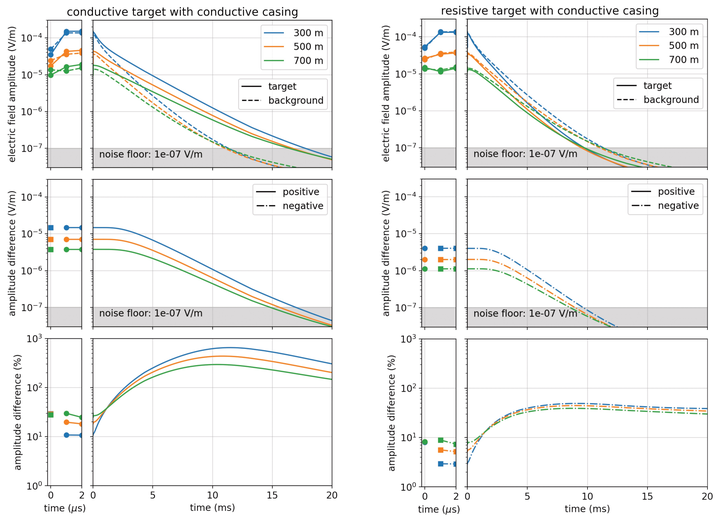
→ galvanic + induced currents contribute to response



Electric field measurements at the surface. To detect target, need:

→ signal above the noise threshold

→ signal to be a significant percentage of primary (no target)



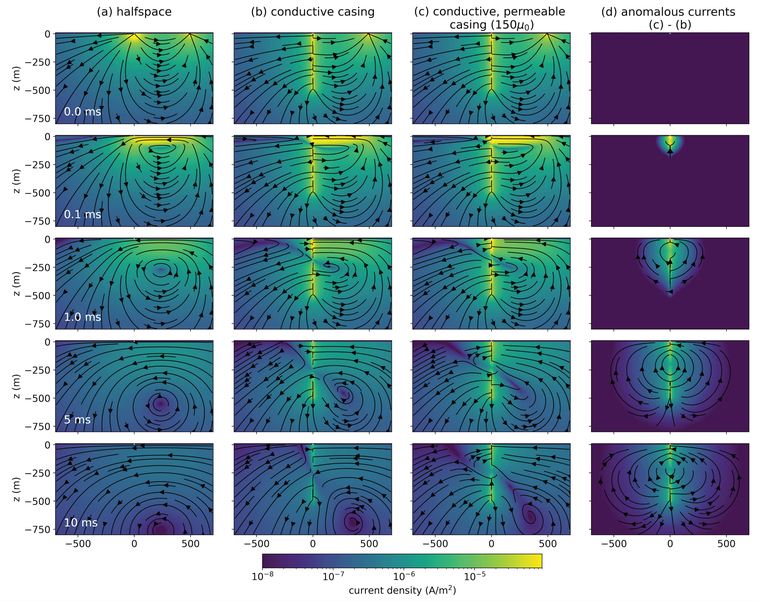
Heagy, L. J., and D. W. Oldenburg. "Electrical and Electromagnetic Responses over Steel-Cased Wells." The Leading Edge 41, no. 2 (2022): 83–92. doi.org/10.1190/tle41020083.1.

Impacts of magnetic permeability

Steel is also magnetic

→ enhances inductive part of the response

→ additional current can enhance excitation of a target



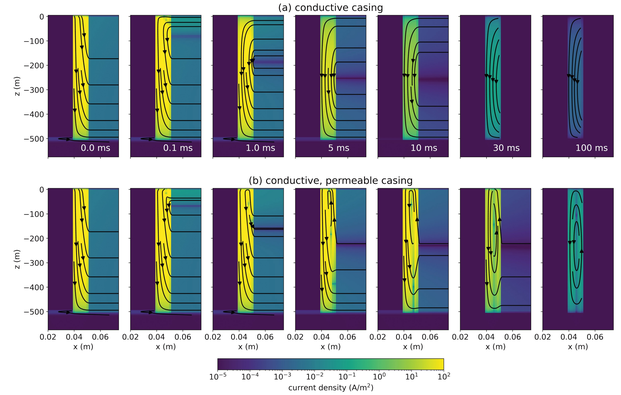
What's happening inside the casing?

Poloidal current system develops in a conductive, permeable casing.

→ enhanced downward-going inductive currents

→ upward-going magnetization current on outer casing wall

$$\nabla \times \vec{b} = \nabla \ln \mu_r \times \vec{b} + \mu \sigma \vec{e}$$



Heagy, L. J., and D. W. Oldenburg. "Impacts of Magnetic Permeability on Electromagnetic Data Collected in Settings with Steel-Cased Wells." arxiv.org/abs/2211.14677.